Samuel Whitebook

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Education

- **California Institute of Technology** — Pasadena, CA
  Doctor of Philosophy in Physics — *Sep 2024 - Present*
- **University of California, Santa Barbara** — Santa Barbara, CA
  Bachelors of Science in Physics | Minor: *Astronomy and Planetary Sciences* — *Sep 2021 - March 2023*
- **College of the Canyons** — Santa Clarita, CA
  Associates of Science in Physics & Associates of Science in Mathematics — *Aug 2018 - May 2021*

Research Experience

- **Discovery of Accreting Brown Dwarf Binaries** — Advisor: Prof. Dimitri Mawet/Prof. Thomas Prince
  California Institute of Technology — *Fall 2024 - Current*
  - **Detection of the First Mass Transferring Brown Dwarf Binaries**: Discovered and followed up the first detected instance of mass transfer between two brown dwarfs.
  - **Development of Substellar Mass Transfer Theory**: Built the theoretical framework for modeling mass transfer in brown dwarfs and low mass stars.
  - **Full Analysis with the James Webb Space Telescope**: Led a successful James Webb Space Telescope proposal as principal investigator to study the atmospheres of mass transferring brown dwarfs.
- **Analysis of ZTF variable and Compact Objects** — Advisor: Prof. Thomas Prince/ Prof. Matthew Graham
  California Institute of Technology — *Summer 2023 - Current*
  - **ZVAR Catalog of Periodic Stars**: Wrote a pipeline to detect periodic sources from the full ZTF catalog of 4.5 billion light curves.
  - **Creation of the FPW Algorithm**: Worked with Thomas Prince and Douglas Finkbeiner to create the Fast Periodicity Weighting (FPW) algorithm, which detects arbitrary waveform periodicity in generic signals, specifically useful for stellar astrophysics.
  - **Catacylsmic Variable Gravitational Waves**: Studied the effects of cataclysmic variable systems on gravitational wave strain in frequencies detectable by NASA's future LISA mission.
  - **Stable Pulsating White Dwarfs**: Discovered candidate stable pulsations in several helium white dwarf stars (DBV) using data from the ZTF catalog.
- **Substellar Companion Detection & Characterization** — Advisor: Prof. Timothy Brandt
  University of California, Santa Barbara — *Spring 2022 - Fall 2024*
  - **KECK NIRSPEC Pipeline**: Coded a data reduction pipeline for KECK II NIRSPEC in high resolution mode in Python.
  - **Discovery of Gliese 229Bb**: Compared the RV of a brown dwarf companion to its main sequence host to detect the existence of a binarity in the first T dwarf, Gliese 229Bab.
  - **SPHERE Instrument Calibration Models**: Used Gaussian process regression to create calibration models for stellar spectral time series taken from the VLT's SPHERE instrument.
  - **Companion Detections with Keck**: Collected direct observation and spectrograph data of brown dwarf companions to main sequence stars with the W.M. Keck Observatory.
- **Deepspace Cosmology, SETI, & Planetary Defense** — Advisor: Prof. Philip Lubin
  University of California, Santa Barbara — *Fall 2021 - Fall 2024*
  - **Local Galactic Transient Detection Software**: Created software to quickly run preprocessing and photometry for a big data pipeline to eventually parse 1π steradian of the sky every 10 seconds.
  - **Dichroic Camera Filter Development**: Managed a group of undergraduates to develop a dichroic camera filter to collect data in two spectral bands at the same time.
  - **Wide Field Telescope Array Design**: Designed an equatorial style telescope mount to hold up to nine wide field telescopes at once.
- **Directed Readings in Accretion Theory** — Advisor: Prof. Omer Blaes
  University of California, Santa Barbara — *Fall 2022*
  - **Advanced Readings in General Relativity**: Studied advanced general relativity specifically related to compact objects.

- **Magnetohydrodynamics**: Learned magnetohydrodynamics to understand the dynamics of accretion.

- **EdgeCube Satellite Control Systems** Advisor: Prof. Garrett Jernigan
  Sonoma State University *Fall 2018 - Fall 2019*
  - **Satellite Control Systems**: Assisted in the design of a 6 degree of freedom control system using momentum transfer devices.
  - **Low Earth Orbit Communications**: Calculated and recorded the projected and official orbital trajectory of the satellite to determine periods of operation.

Teaching and Work Experience

- **Undergraduate Teaching Assistant**
  University of California, Santa Barbara *Fall 2022 - Spring 2023*
  - **Physics 134L, Observational Astrophysics (2 times)**: Worked with Professor Philip Lubin as an undergraduate teaching assistant for UCSB's upper division observational astrophysics lab course.
  - **Physics 29, Introduction to Programming (2 times)**: Worked as an undergraduate teaching assistant for UCSB's introduction to programming for physics course.

- **Undergraduate Remote Laboratory Maintenance** Advisor: Prof. Deborah Fygenson
  University of California, Santa Barbara *Winter 2021 - Summer 2022*
  - **Atomic Spectroscopy Circuitry**: Developed custom circuitry to assist in the remote control of an atomic spectroscopy lab for second year undergraduate students.
  - **Remote Lab Troubleshooting and Support**: Assisted several classes of students operating remote labs. Answered questions and helped fix various apparatuses.
  - **Custom Component Design**: Designed custom components to improve the design of several labs.

- **Professional Tutor** Supervisor: Suzanne Reed
  Mathnasium of Canyon Country *Fall 2020 - Spring 2023*
  - **K-12 and lower division college Math and Physics Tutor**: Taught mathematics and physics to students ranging in age from kindergarten to second year college students. Some courses tutored include single and multivariable calculus, linear algebra, and differential equations.

Selected Publications

- *The Physics of Mass Transfer in Substellar and Low-Mass Binaries* | 1st Author *June 2026* **ApJL, 1004, L39**
- *A Mass Transferring Brown Dwarf Binary on a 57 Minute Orbit* | 1st Author *March 2026* **ApJL, 1000, L23**
- *A Fast Periodicity Detection Algorithm Sensitive to Arbitrary Waveforms* | 3rd Author *May 2025* **PASP, 137, 054504**
- *Discovery of the Binarity of Gliese 229B, and Constraints on the System's Properties* | 1st Author *October 2024* **ApJL, 974, L30**

Presentations and Talks

- **Samuel Whitebook**. *The Discovery of Stable Substellar Mass Transfer and ZTFJ1239+8347.* Department of Astronomy and Astrophysics, University of California Berkeley, April 23th, 2026.
- **Samuel Whitebook**. *Strange Giants: High Mass Brown Dwarfs and Gliese 229B.* Astronomy on Tap, Las Cumbres Observatory, Santa Barbara, July 6th, 2023.
- **Samuel Whitebook**. *Detection and Characterization of Substellar Companions.* Kavli Institute for Theoretical Physics, University of California, Santa Barbara, September 9th, 2022.

Observing Experience

- **James Webb Space Telescope** | NIRSpec
- **200 in. Hale Telescope** | DBSP/NGPS, Palomar Observatory
- **200 in. Hale Telescope** | CHIMERA, Palomar Observatory
- **200 in. Hale Telescope** | WIRC, Palomar Observatory
- **Subaru** | CHARIS, National Astronomical Observatory of Japan
- **Keck I** | LRIS, W.M. Keck Observatory
- **Keck II** | ESI, W.M. Keck Observatory
- **Keck II** | NIRC2, W.M. Keck Observatory
- **Keck II** | NIRES, W.M. Keck Observatory
- **Keck II** | NIRSPEC, W.M. Keck Observatory

Activities

- **Caltech Astronomy Outreach**: Actively volunteering for several outreach programs at Caltech including stargazing lectures and Astronomy on Tap
- **Leader of AstroTea Journal Club**: Organized the astronomy department's research journal club managing paper choices and presentation volunteers from the beginning of fall 2022 until summer 2023
- **UCSB Transfer Peer Mentor**: Mentored two new transfers to UCSB through the peer mentorship program. Assisted students with picking classes, learning studying strategies, and integrating into the university community.
- **Letters to a Pre-Scientist Pen Pal**: Corresponded with an elementary school student to discuss the work of a STEM researcher and promote engagement with underrepresented communities.

Honors and Awards

- James A. Cullen Memorial Graduate Fellowship | Caltech
- Harold Brown Graduate Fellowship | Caltech
- Dean's Honors in the College of Letters and Sciences | UCSB
- Transfer Student Research Award | UCSB
- NVIDIA Applied Research Accelerator Award | UCSB

Skills

- **Mathematics**: Very strong grasp of advanced statistics such as periodograms, Gaussian process regression, and principal component analysis
- **Languages**: Python (tensorflow, astropy, scipy), C++, CUDA, C#, Wolfram Mathematica
- **Physical Skills**: soldering, welding, machining, private pilot, commercial telescope maintenance & operation